Aivura

CURSOR BUILDATHON - PROJECT SUBMISSION DOCUMENT

Cursor x TechTalk360 - Best Use of n8n

Repository: github.com/Kiruthiyan/n8n_aivura | Bot: @Aivura_bot

Project	Aivura - AI Academic Co-pilot on Telegram
Team Name	Cyber League
Track	Best Use of n8n
Demo URL	https://t.me/Aivura_bot
Repository	https://github.com/Kiruthiyan/n8n_aivura

Section 01 - Project Overview

Project Name & One-Line Pitch

Aivura - An AI academic co-pilot on Telegram that turns lecture PDFs, deadlines, and study questions into structured, actionable replies for university students. The product is orchestrated entirely in n8n and built with Cursor.

Summary

University students often manage assignments, exams, emails, Drive files, and study notes across disconnected tools. This causes missed deadlines, repeated manual work, and wasted study time. Aivura solves this by giving students one familiar interface: Telegram. Students can message the bot, upload PDFs, or run slash commands, while n8n verifies access, classifies the intent, routes the request to the correct workflow, and returns a consistent AI-generated response. Admins manage student access through a Cursor-built Admin Portal and a Notion user registry, without requiring student signup or a separate mobile app.

Submission Details

Item	Detail
Track	Best Use of n8n
Track integration	12 n8n workflows: 1 router workflow and 11 feature sub-workflows. The build uses Telegram Trigger, Execute Workflow, Switch routing, HTTP nodes for Gemini / Notion / Gmail / Telegram File API, Code nodes for authentication, PDF vision, and demo mode.
Cursor integration	The full repository was scaffolded and iterated in Cursor, including the workflow JSON generator, React Admin Portal, documentation, and security patterns.
Team name	Aivura Team [update if your official team name is different]
Demo URL	https://t.me/Aivura_bot
Repository	https://github.com/Kiruthiyan/n8n_aivura

Section 02 - Problem Statement

The Problem

Who: Undergraduate and postgraduate students at Sri Lankan and regional universities, especially students in technology and business faculties with heavy assignment loads.

Context: During semester peaks, students receive assignment briefs by email, store files in Drive, track tasks in Notion or WhatsApp groups, and ask study questions in separate AI tools. These systems are not connected to their own documents, deadlines, or academic workflow.

Consequence if unsolved: Students may miss deadlines, revise inefficiently, create inconsistent assignment structures, and experience unnecessary stress during exam periods.

Why It Matters

Dimension	Detail
Frequency	Daily during the semester, with multiple academic events per week per student.
Severity	Direct impact on grades, revision quality, confidence, and student workload management.
Current workaround	Generic AI chat tools, manual Notion boards, WhatsApp study groups, and repeated searching through emails or PDFs.
Why workarounds fail	They do not provide one authenticated channel connected to the student’s real academic data, documents, deadlines, and tasks.

Root Cause

The root cause is not the lack of AI models. It is the fragmentation of academic workflows across communication channels, file storage, emails, task boards, and study tools. Students need an orchestration layer that connects these tools and delivers results through the mobile channel they already use.

Section 03 - Proposed Solution

What the Product Does

Aivura is a Telegram-native student assistant. A student can message @Aivura_bot or upload a PDF with a caption. n8n receives the webhook, verifies the Telegram Chat ID through Notion or demo allowlist, detects the user intent, and executes exactly one specialized sub-workflow. Each sub-workflow gathers the required context, calls Gemini 2.0 Flash with a feature-specific prompt, formats a structured Markdown reply, and sends it back to Telegram.

Core mechanism: Workflow-orchestrated AI. Business logic, access control, routing, context fetching, and integrations live in n8n. The LLM is used as one controlled step inside deterministic workflows, rather than as an unstructured black-box chat UI.

Key Features Built and Demonstrable

Feature	User need addressed
PDF / document summary	Turn long readings into exam-ready points without manually re-reading everything.
MCQ generation	Self-test before quizzes using any topic or notes.
Submission answer draft	Create a structured assignment outline as a study aid for the student to review and verify.
Study plan	Generate time-boxed study blocks using tasks and deadlines when integrated.
Deadlines digest	Surface urgent academic items from email and Notion.
Viva preparation	Generate likely questions and model answers before oral exams.
Lecture summary	Compress lecture notes or PDFs for revision.
Flashcards	Create revision cards from any topic.
Today plan	Plan the next few hours of study work.
Explain simple	Explain a difficult topic in plain language using /simple.
General ask / help	Provide tutoring support and command discovery.
Admin Portal	Add students, export user data for Notion, and support broadcast messages.
Access gate	Deny unregistered users and allow demo mode for hackathon testing.
Gemini Vision PDF	Extract information from PDFs or images for summary workflows.

Scope

In scope for this build:

- Telegram bot for messages and documents
- 12 n8n workflow files: 1 router and 11 student feature workflows
- Notion-based user registry with demo mode bypass
- Google Gemini 2.0 Flash for intent detection, responses, and PDF vision
- Optional Gmail and Notion student database context for richer flows
- Cursor-built Admin Portal, ready for local or Vercel deployment
- Workflow generator for reproducible n8n JSON exports

Deliberately out of scope for this build:

- Full Sinhala and Tamil localization
- Student self-registration and payments
- Multi-university tenancy and billing
- Native mobile app, because Telegram is used as the client

Section 04 - Functional Requirements

Must Have

ID	Requirement
M1	Telegram Trigger receives text messages and documents.
M2	The system rejects unregistered or inactive Chat IDs with a clear message.
M3	Registered users can receive a /help command list.
M4	PDF upload triggers the document-summary workflow.
M5	Slash commands map to the correct intent, such as /mcq, /studyplan, and /deadlines.
M6	Gemini returns a consistent four-section Markdown response: Summary, Findings, Actions, and Draft/Plan.
M7	The router Switch executes only one sub-workflow per request.
M8	Admin can add a student and Chat ID in the portal and export the record for Notion.
M9	All secrets are loaded through environment variables in n8n, with no API keys stored in workflow JSON.

Should Have

ID	Requirement
S1	Natural-language intent detection when no slash command is used.
S2	Gmail fetch for deadline and study-plan workflows.
S3	Notion student database query for tasks and deadlines.
S4	Telegram file download and Gemini Vision for PDFs/images.
S5	continueOnFail and aggregated api_errors in workflow context.
S6	Demo mode using AIVURA_DEMO_MODE for hackathon testing without Notion.
S7	Admin broadcast through the portal or CLI script.

Could Have / Won't Have in This Build

ID	Item	Decision
C1	Google Drive file listing	Won't have - Gmail and Notion are sufficient for the demo.
C2	Voice messages	Won't have - not required for the core workflow demonstration.
C3	Sinhala/Tamil full localization	Could have - planned after the hackathon.
C4	Usage analytics dashboard	Could have - the portal currently includes an Activity stub.
C5	Automated Notion sync from portal	Won't have - manual export is more reliable for the hackathon build.

Section 05 - Non-Functional Requirements

Performance

Interaction	Target
Text command without Gmail	5-15 seconds end-to-end, mainly dependent on Gemini response time.
Study plan / deadlines with Gmail and Notion	15-30 seconds.
PDF summary	10-25 seconds, including download and AI processing.
Concurrent users	n8n queue supports a modest campus pilot; scaling can be handled with n8n workers and Gemini rate-limit planning.

Reliability & Error Handling

- HTTP nodes use continueOnFail and alwaysOutputData where appropriate.
- Code nodes collect api_errors and pass them to Gemini for graceful degradation.
- Denied users receive a fixed message without stack traces.
- Sub-workflow failures route to a collect-reply step with fallback text.

Usability

- Students use Telegram only, with no extra onboarding UI.
- Commands are discoverable through /help.
- Admins use one portal for user management and workflow documentation.
- Responses follow a predictable four-section format for easier reading on mobile.
- The portal uses semantic HTML and a contrast-friendly dark theme.

Scalability

Layer	Scale path
n8n	Use horizontal workers and separate production/staging instances.
Workflows	Keep features isolated so high-demand paths such as PDF summary can scale independently.
Notion	Use per-cohort databases and pagination on queries.
Gemini	Upgrade API tier and add caching for repeated summaries.

Section 06 - Technical Architecture

System Overview

The architecture connects Telegram, n8n, Gemini, Notion, Gmail, and the Admin Portal. Telegram is the user interface. n8n is the orchestration layer. Gemini handles classification, reasoning, response generation, and document vision. Notion stores the user registry and optional student tasks. Gmail provides academic email context for deadline and study-plan flows. The Admin Portal supports student access management and manual export.

Simplified flow: Telegram student request -> n8n Router Workflow -> access check -> intent detection -> Switch routing -> selected feature sub-workflow -> Gemini response -> Telegram reply.

Technology Stack

Layer	Technology	Rationale
Orchestration	n8n	Visual, exportable workflows with Telegram, HTTP, Code, Switch, and

		Execute Workflow nodes.
AI	Google Gemini 2.0 Flash	Fast and cost-effective, with JSON mode for intent detection and vision support for PDFs/images.
Messaging	Telegram Bot API	Widely available on student phones and supports file upload.
User registry	Notion API	Low-friction admin database with filtering by Chat ID.
Tasks / context	Notion Student DB	Structured deadline and task context for study flows.
Email context	Gmail API	Academic keyword search for deadline and study-plan support.
Admin UI	React 18, Vite, TypeScript, Tailwind	Built rapidly in Cursor and deployable to Vercel.
Workflow codegen	Node.js script	Keeps similar workflow JSON files maintainable and reproducible.
IDE / build	Cursor	Primary development environment for code, workflow scaffolding, and documentation.
Secrets	n8n environment variables	Prevents keys from being committed to the repository or workflow exports.

Data Flow: Student Sends a PDF

1. Telegram sends the request to the n8n Telegram Trigger.
2. Extract node reads chat_id, user_message, file_id, and mime_type.
3. Check Registered node queries Notion for the Telegram Chat ID.
4. If the user is inactive or not found, the deny message is sent and the workflow ends.
5. Detect Intent calls Gemini in JSON mode to classify the request.
6. Parse Intent applies command overrides and routes PDF uploads to DOCUMENT_SUMMARY.
7. Switch node executes the PDF summary sub-workflow.
8. The sub-workflow downloads the file through the Telegram API, extracts text using Gemini Vision where needed, builds context, calls Gemini, and formats the reply.
9. Collect Reply gathers the final output.
10. Telegram Reply sends the structured Markdown response to the student.

AI Integration

Use	Model / Method	Prompting role
Intent classification	Gemini 2.0 Flash with application/json response	Classifies the request into one of the supported intents using message text and file presence.
Feature responses	Gemini 2.0 Flash, temperature 0.4	Uses a feature-specific system prompt and JSON user payload containing message, context, and errors.
PDF extraction	Gemini Vision inline data	Extracts readable text from PDFs or images before summarization.
Output contract	Fixed Markdown sections	Keeps replies readable and consistent on mobile.

Known Technical Limitations

- The portal-to-Notion sync is manual for hackathon reliability; automated sync is planned later.
- Admin authentication uses an environment-based password for the hackathon build; production should use server-side authentication or SSO.
- The system currently has no persistent chat memory across sessions; each message is treated as stateless.
- PDF processing depends on Gemini Vision and fallback handling if binary extraction fails or API limits are reached.

Section 07 - Security

Authentication & Authorisation

Actor	Mechanism
Student	Telegram Chat ID is matched in Notion, and the Active checkbox must be true.
Admin	Admin Portal password through VITE_ADMIN_PASSWORD, with a localStorage session for the hackathon build.
n8n operator	n8n instance login is required and documented in the security notes.

There is no public student login surface. This reduces the attack area and keeps access controlled through Telegram Chat ID verification.

Data Handling

- Messages pass through n8n execution logs, so the n8n instance must be secured.
- Notion stores registration metadata such as name, Chat ID, role, and active status.
- Gmail fetches limited metadata and snippets for context where enabled.
- No student passwords are stored.

API & Secret Management

- The repository includes .env.example placeholders only.
- The .gitignore prevents .env files from being committed.
- Workflow JSON files use environment variables such as GEMINI_API_KEY and TELEGRAM_BOT_TOKEN.
- No production API keys are intentionally stored in the repository or workflow exports.

Input Validation

- Chat IDs are coerced to the expected type before Notion filtering.
- Slash commands are parsed using a whitelist map.
- Telegram replies are truncated to remain within Telegram message limits.
- continueOnFail reduces the risk of one API failure crashing the entire pipeline.

Known Vulnerabilities or Gaps

Gap	Mitigation plan
Admin password is stored in client-side environment for hackathon build	Move to server-side authentication or SSO after the hackathon.
No full rate limiting yet	Add n8n throttle controls and Gemini quota management.
Telegram bot can be messaged by anyone	Use Notion gate and optional allowlists per cohort.
Execution logs may contain personal data	Apply retention policy and restrict hosted n8n access.

Section 08 - User Stories & Use Cases

Core User Stories

- As a student, I want to send my lecture PDF to Telegram, so that I get a structured summary before class.
- As a student, I want to run /mcq on a topic, so that I can self-test without manually writing questions.
- As a student, I want to run /deadlines, so that I can see urgent items from email and my task board in one message.
- As a student, I want to run /studyplan before finals, so that I get a realistic daily study schedule.
- As a student, I want to run /viva, so that I can practise likely oral exam questions.
- As an admin, I want to register a student by Chat ID, so that only my cohort can use the bot.
- As an admin, I want to deactivate a user, so that access is revoked without deleting history.
- As a student, I want to run /help, so that I know all available commands.
- As a judge, I want to see n8n route requests to different workflows, so that the orchestration depth is clear.

Primary Use Case Walkthrough

Step	User action	System action
1	Admin adds a student name and Chat ID in the portal, exports the record, and sets Active in Notion.	Notion row is created or updated.
2	Student opens Telegram and sends /help.	Router checks access and returns the command list.
3	Student uploads OS_Chapter3.pdf with the caption “Summarize for exam”.	Router sends the request to the PDF workflow. Gemini Vision extracts content and Gemini generates the summary.
4	Student reads the reply.	Telegram shows the four-section Markdown response.
5	Student sends /studyplan exams in 5 days.	Study-plan workflow uses available Gmail and Notion context to return a personalized plan.

Edge Cases

Case	Behaviour
Unregistered Chat ID	Polite deny message with instruction to contact the admin.
Inactive user	Same deny path as an unregistered user.
Message with no text, only PDF	Intent defaults to DOCUMENT_SUMMARY.
Gmail token missing	Workflow continues and the AI notes that email context is unavailable.
Gemini timeout or quota issue	Fallback four-section error template is returned.
Unknown slash command	AI intent detection runs and falls back to GENERAL_STUDY where appropriate.

Section 09 - Target Users & Market

Primary User

The primary users are university students, especially students in Sri Lanka and similar markets who rely heavily on smartphones and messaging apps. The first target group is students in English-medium technical and business programs with frequent assignments, exams, and PDF-based learning material.

Market Opportunity

- Sri Lanka has a large university student population, especially in STEM and business fields.
- The first market wedge is a department-level pilot: one faculty, one bot, and one admin-managed roster.
- The total opportunity grows through institutional licenses instead of relying only on individual student subscriptions.

Competitive Landscape

Alternative	Weakness vs. Aivura
Raw ChatGPT	No personal deadlines, no workflow routing, no Telegram access gate, and no institution-controlled roster.
Notion AI	Not Telegram-native, and many students do not use Notion as their main mobile study interface.
Generic study apps	Require another app and usually do not integrate with the student’s real email, tasks, and documents.

Aivura’s wedge is an orchestrated, authenticated, mobile-first academic agent. n8n acts as the integration layer that connects messaging, access control, APIs, and AI.

Section 10 - Business Model

Revenue Model

Aivura is best suited to a B2B2C model through universities, departments, and private institutes. Institutions can license the bot for a cohort of students and receive admin access, support, and workflow configuration. This model fits the product because students may have low willingness to pay individually, while departments benefit from better student support and engagement.

Pricing Hypothesis

Tier	Price hypothesis	Includes
Pilot	Free for 3 months	One department and up to 100 students.
Department	Approximately LKR 150,000-300,000 per year	Up to 500 students, bot access, admin portal, and optional Gmail context.
Campus	Custom pricing	SSO, analytics, Sinhala/Tamil support, and compliance package.

Go-to-Market

1. Acquire the first 10 customers through direct outreach to department heads and student union technology representatives at SLIIT, Moratuwa, Colombo, NSBM, and similar institutions.
2. Use the Buildathon demo, LinkedIn build-in-public posts, and faculty WhatsApp/Telegram communities as the main awareness channels.
3. Lead with the message: “Your students already use Telegram - give them an AI assistant you control.”

Roadmap

Horizon	Deliverable
Now - hackathon	12 workflows, Admin Portal, broadcast support, demo mode, 11+ commands, and Gemini Vision.
0-3 months	Improved PDF OCR, automatic Notion sync, usage dashboard, and Sinhala/Tamil support.
12 months	Multi-tenant admin, Moodle/LMS integration, institutional billing, analytics, and compliance package.

Section 11 - Why This Project Leads the Track

Technical Edge

- This is not a single chatbot workflow. It uses a production-style router and 11 Execute Workflow sub-flows.
- It includes real integrations: Telegram Trigger, Notion authorization, Gmail API, Telegram File API, and Gemini for vision, intent, and response generation.
- Switch node routing makes the execution graph visible, debuggable, and judge-friendly.
- Workflow-as-code through the generator script shows maintainability beyond manual drag-and-drop.
- Failure-aware orchestration uses error aggregation and graceful AI fallbacks.

Problem-Solution Fit

Students do not need another app. They need connected academic automation inside a communication tool they already use. Aivura directly connects the problem of fragmented student workflows with the solution of n8n-powered orchestration on Telegram.

Execution Quality

- End-to-end demo path is documented.
- A separate test matrix exists for Telegram workflow testing.
- Security and architecture notes are included.
- Both the admin journey and student journey are functional.

Real-World Potential

Aivura is pilot-ready for a single department. It has a clear monetization path, a practical expansion roadmap, and a technical route from hackathon workflow JSON to hosted n8n Cloud and a deployed Vercel Admin Portal.

Section 12 - Team & Roles

Team Members

Name	Role	Built
[Name 1]	Lead / n8n Architect	Router workflow, sub-workflows, Notion/Gmail integration, and generator script.
[Name 2]	Frontend / Cursor	Admin Portal in React, user experience, and deployment setup.
[Name 3]	AI / Product	Prompts, intent design, demo script, and submission document.

Note: Replace the placeholder names with your actual team member names before final submission.

Why This Team

This team combines workflow automation, rapid UI development with Cursor and React, AI prompt design, and real student-domain understanding. That combination makes the team capable of shipping a full product experience: Telegram bot, admin portal, AI workflows, integrations, and documentation within hackathon time.

Appendix - Demo Checklist

1. Open the n8n canvas and show the Router with Execute Workflow branches.

2. Open the Admin Portal and add a student with Chat ID.

3. Show an unregistered Telegram user being denied access.

4. Send /help from a registered Telegram account.

5. Upload a PDF and show the generated summary.

6. Run /mcq binary trees and show the response.

7. Open a successful n8n execution and show the active nodes in the workflow graph.

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